

EXPERIMENT-4

AIM: To determine the surface tension of given liquids with the stalagmometer.

Apparatus required: Stalagmometer, Beaker, specific gravity bottle

Chemical required: Distilled water, Unknown liquid

Principle:

Surface tension of liquids

The molecules of liquids attract each other by cohesive forces resulting into small distances between the molecules (on the order of 0.1 nm). Thus the compressibility of liquids is lower than that of gas, while the density is much higher. On the other hand, these cohesive forces are not strong enough to result into the fixed position of molecules that can be seen in solid matter. Liquids do not keep a fixed shape, but adapt the shape of a container. Attractive cohesive forces are short-range forces, which are based on the electronic interactions. They affect molecules in their close vicinity only (zone of molecular interaction). In the bulk of the liquid, each molecule is attracted equally in all directions by the neighbouring molecules, hence zero net force (Fig. 1). However, the molecules at the surface do not have other like molecules on all sides around them and they are pulled inwards the liquid core by non-zero net force Fig. 1). Consequently, they cohere more strongly to those associated with them directly on the surface and form a surface “film”. Nevertheless, these surface molecules are in the energetically unfavourable state which forces liquid to minimize the surface area. The geometrical requirement of smallest surface area at the fixed volume is satisfied by the sphere. It is the reason why the free drops of water form spherical droplets.

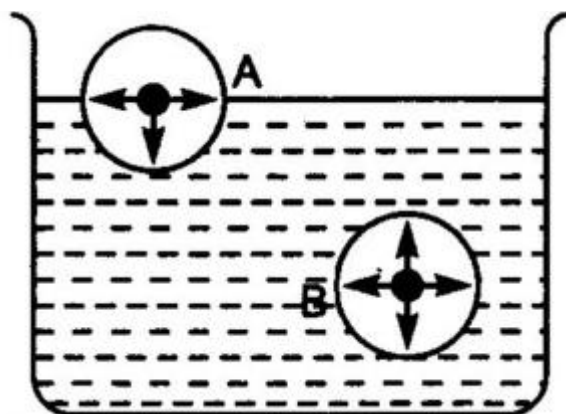


Figure 1: Cohesive forces in a case of the molecule at the liquid surface and in the bulk.

Surface area increase, S , is balanced out by the increase of energy, E , where

$$\frac{E}{S} \quad (1)$$

The coefficient defines the **surface tension** in the units of $\text{J/m}^2 = \text{N/m}$. Each liquid is characterized by its own surface tension, which decreases with an increasing temperature (See. Table 1).

The determination of the surface tension by the drop counting method

There are two line marks on the stalagmometer: top line above the wide part and bottom line below it. The volume between these two lines is V , and liquid with density ρ contained in this volume has a mass m

$$m = V \cdot \rho \quad (8)$$

Such a volume V corresponds to n drops, which are released from the stalagmometer upon the decrease of liquid level from top to bottom line mark. Here, the average mass of one drop

$$\frac{m}{n} = \frac{V \cdot \rho}{n} \quad (9)$$

Substituting the mass m in the equation (7) with the mass of one drop then yields

$$\sigma = \sigma_{\text{H}_2\text{O}} \cdot \frac{\rho}{\rho_{\text{H}_2\text{O}}} \cdot \frac{n_{\text{H}_2\text{O}}}{n} \quad (10)$$

U

Procedure

1. Fill the stalagmometer up to the top mark with distilled water. Release water to the weighing bottle and count how many drops it takes to decrease the water level in stalagmometer down to bottom mark. Write down the number of drops $n_{\text{H}_2\text{O}}$.
 2. Empty and dry the weighing bottle and stalagmometer, and prepare them for the next measurement.
 3. Repeat steps 1 and 2 for liquids with the unknown surface tension.
 4. Write down the densities of studied liquids according to the notes on bottles, and density of distilled water at the actual temperature in laboratory from the table
1. Using equation (10), calculate the surface tension for all the studied liquids.

Observations:

Room temp. = ... °C

Table I.

The determination of the surface tension by the drop counting method

S.No.	Water		Given Liquid	
	Number of drops	Mean	Number of drops	Mean
1.				
2.				
3.				

Weight of empty specific gravity bottle = w_1 (15.627 g)
 Weight of empty specific gravity bottle with water = w_2 (26.813 g)
 Weight of empty specific gravity bottle with liquid = w_3 (26.770 g)
 Density of water at 25 °C = 72.14 dynes/cm
 $\therefore$ Weight of water = $(w_2 - w_1)$ g
 Weight of liquid = $(w_3 - w_1)$ g
 Relative density of the liquid = $d_l / d_w = w_3 - w_1 / w_2 - w_1$
 $\gamma_l / \gamma_w = d_l / d_w \times n_w / n_l$

where, γ_l , d_l , n_l = surface tension, density and number of drops of liquids.

γ_w , d_w , n_w = surface tension, density and number of drops of water.

By putting the respective values, the surface tension of the liquid relative to water is found out.

Absolute surface tension of the liquid. $\gamma_l = n_w / n_l \times d_l / d_w \times \gamma_w = \dots\dots\dots \text{Nm}^{-1}$.

Result: The surface tension of the given liquid is.....dyne/cm

Viva-Voce

- Q1. What are the effects of Surface tension?
- Q2. Give some application of Surface Tension.
- Q3. What is the procedure to measure surface tension by stalagmometer?
- Q4. Why we use narrow capillary in the stalagmometer?
- Q5. Define surface energy.
- Q6. Explain 'the drops of liquid or bubbles of a gas are spherical in shape'.